Supplementary material

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Proof of Lemma 1: Fix $i \in \mathcal{V}$ and $t \in \{0, \dots, N-k\}$. Iterating (2), for $j = 0, \dots, k-1$,

$$\theta_i(t+j) = \theta_i(t) + \underbrace{(A^j - I)\theta_i(t) + \sum_{l=0}^{j-1} A^{j-1-l} w_i(t+l)}_{\Delta_{i,t}(j)}, \tag{S1}$$

so the stacked measurements of the window satisfy:

$$\begin{aligned} \bar{y}_{i,t} &= \bar{C}_{i,t}\theta_i(t) + \bar{b}_{i,t} + \bar{w}_{m,i,t}, \\ \bar{b}_{i,t} &:= \operatorname{col}\big(C_{i,t+j}\Delta_{i,t}(j)\big)_{j=0}^{k-1}, \qquad \bar{w}_{m,i,t} \sim \mathcal{N}(0, \bar{R}_i). \end{aligned} \tag{S2}$$

(i) Excitation: The window $\{t, \dots, t+k-1\}$ contains the $\lfloor k/k_0 \rfloor$ disjoint blocks $\{t + qk_0, \dots, t + (q+1)k_0 - 1\}$, $q = 0, \dots, \lfloor k/k_0 \rfloor - 1$, each starting at a time $t + qk_0 \le N - k_0$, so that (A3) applies to each of them. With $\bar{R}_i^{-1} = I_k \otimes R_i^{-1} \succeq \lambda_{\max}(R_i)^{-1} I_{kn}$,

$$\bar{C}_{i,t}^{\mathsf{T}} \bar{R}_i^{-1} \bar{C}_{i,t} \succeq \frac{1}{\lambda_{\max}(R_i)} \sum_{q=0}^{\lfloor k/k_0 \rfloor - 1} \sum_{j=0}^{k_0-1} C_{i,t+qk_0+j}^{\mathsf{T}} C_{i,t+qk_0+j} \succeq \frac{\lfloor k/k_0 \rfloor \alpha_0}{\lambda_{\max}(R_i)} I_p = \alpha_{k,i} I_p, \tag{S3}$$

which is the first claim.

(ii) Noise and bias: $\eta_i(t)$ is a zero-mean Gaussian function of $w_{m,i}(t), \dots, w_{m,i}(t+k-1)$ only, and, by (S1), $\beta_i(t)$ is a zero-mean linear function of $\theta_i(t), w_i(t), \dots, w_i(t+k-2)$ only. As these variables are independent, $\eta_i(t)$ is independent of every $\beta_i(s)$, and of $\eta_i(s)$ whenever $|t-s| \ge k$; and the noises and biases of different agents are independent. By (S3), $\bar{R}_i \succeq \lambda_{\min}(R_i) I_{kn}$, and $\lfloor k/k_0 \rfloor \ge k/(2k_0)$,

$$\begin{aligned} \Sigma_{\eta,i}(t) &= K_{i,t} \bar{R}_i K_{i,t}^{\mathsf{T}} = \big(\bar{C}_{i,t}^{\mathsf{T}} \bar{R}_i^{-1} \bar{C}_{i,t}\big)^{-1} \preceq \alpha_{k,i}^{-1} I_p, \\ \|K_{i,t}\|^2 &\le \frac{\|K_{i,t} \bar{R}_i K_{i,t}^{\mathsf{T}}\|}{\lambda_{\min}(R_i)} \le \frac{\lambda_{\max}(R_i)}{\lfloor k/k_0 \rfloor \, \alpha_0 \, \lambda_{\min}(R_i)} \le \frac{\chi_i^2}{k}. \end{aligned} \tag{S4}$$

The first relation is the second claim.

(iii) Bias: $\theta_i(t)$ is independent of the later zero-mean noises. Moreover, $\|A^j - I\| = \|(A - I)\sum_{l<j} A^l\| \le C_\rho \, j \|A - I\|$ and $\mathbb{E}\|\sum_{l<j} A^{j-1-l} w_i(t+l)\|^2 = \sum_{l<j} \operatorname{tr}\big(A^{j-1-l} Q_i (A^{j-1-l})^{\mathsf{T}}\big) \le C_\rho^2 \, j \operatorname{tr} Q_i$. Hence $\mathbb{E}\|\Delta_{i,t}(j)\|^2 \le C_\rho^2 \big(j^2 \|A - I\|^2 B_{\theta,i}^2 + j \operatorname{tr} Q_i\big)$, and with $\|C_{i,t+j}\| \le \Gamma_i$, $\sum_{j<k} j^2 \le k^3/3$, and $\sum_{j<k} j \le k^2/2$,

$$\mathbb{E}\|\bar{b}_{i,t}\|^2 = \sum_{j=1}^{k-1} \mathbb{E}\|C_{i,t+j}\Delta_{i,t}(j)\|^2 \le \Gamma_i^2 C_\rho^2 \Big(\frac{k^3}{3}\|A - I\|^2 B_{\theta,i}^2 + \frac{k^2}{2} \operatorname{tr} Q_i\Big). \tag{S5}$$

By (S4), (S5), and $\sqrt{a+b} \le \sqrt{a} + \sqrt{b}$,

$$\big(\mathbb{E}\|\beta_i(t)\|^2\big)^{1/2} \le \|K_{i,t}\| \big(\mathbb{E}\|\bar{b}_{i,t}\|^2\big)^{1/2} \le \frac{\chi_i}{\sqrt{k}} \Gamma_i C_\rho \Big(\frac{k^{3/2}}{\sqrt{3}}\|A - I\| B_{\theta,i} + \frac{k}{\sqrt{2}} \sqrt{\operatorname{tr} Q_i}\Big) = b_{k,i}. \tag{S6}$$

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(iv) Conclusion: By (6) and Minkowski's inequality,

$$\begin{aligned}\big(\mathbb{E}\|\tilde{\theta}_i(t)-\theta_i(t)\|^2\big)^{1/2} &\le \big(\mathbb{E}\|\eta_i(t)\|^2\big)^{1/2}+\big(\mathbb{E}\|\beta_i(t)\|^2\big)^{1/2} \le \sqrt{p/\alpha_{k,i}}+b_{k,i}=\epsilon_i^{\mathrm{w}},\\ \big(\mathbb{E}\|\tilde{\theta}_i(t)\|^2\big)^{1/2} &\le \big(\mathbb{E}\|\theta_i(t)\|^2\big)^{1/2}+\big(\mathbb{E}\|\tilde{\theta}_i(t)-\theta_i(t)\|^2\big)^{1/2} \le B_{\theta,i}+\epsilon_i^{\mathrm{w}} =: B'_{\theta,i}.\end{aligned} \tag{S7}$$

The first bound is (14) and the second is used in the proofs of Theorems 2 and 4.

Proof of Theorem 2: The proof has five steps.

(i) Covariance of the window estimates: Let $\tilde{\Theta}_i := \mathrm{col}(\tilde{\theta}_i(0),\dots,\tilde{\theta}_i(N-k))$. The exploration points, and hence the gains $K_{i,t}$, are fixed in advance, so $\tilde{\Theta}_i$ is a linear function of the measurements $\tilde{y}_i(0),\dots,\tilde{y}_i(N-1)$. Since the measurements are zero-mean Gaussian, so is $\tilde{\Theta}_i$, and $\|\mathbb{E}[\tilde{\Theta}_i\tilde{\Theta}_i^{\mathsf{T}}]\|$ is at most χ_i^2 times the norm of the covariance of the measurements (the linear map from the measurements to $\tilde{\Theta}_i$ has norm at most χ_i, since each measurement enters at most k of the windows and $\|K_{i,t}\|^2 \le \chi_i^2/k$ by (S4)). That covariance is the sum of a noise part, of norm $\lambda_{\max}(R_i)$, and a parameter part, of norm at most Γ_i^2 times that of the covariance of $\mathrm{col}(\theta_i(0),\dots,\theta_i(N-1))$. The blocks $\mathbb{E}[\theta_i(t)\theta_i(s)^{\mathsf{T}}] = A^{t-s}P_i(s)$, $t\ge s$, have norm at most $C_\rho B_{\theta,i}^2\rho_A^{t-s}$, and a symmetric block matrix has norm at most the largest sum of the block norms along a block row, here at most $2C_\rho B_{\theta,i}^2/(1-\rho_A)$. Therefore

$$\|\mathbb{E}[\tilde{\Theta}_i\tilde{\Theta}_i^{\mathsf{T}}]\| \le c_1 := \max_{l\in\mathcal{V}} \chi_l^2\Big(\frac{2C_\rho\Gamma_l^2B_{\theta,l}^2}{1-\rho_A}+\lambda_{\max}(R_l)\Big). \tag{S8}$$

(ii) Concentration of the moments: An entry (a,b) in (8) is $(\hat{M}_{j,i})_{ab} = \tilde{\Theta}_i^{\mathsf{T}}\Pi\,\tilde{\Theta}_i/(N-2k)$, where the matrix Π has, for each $t=k,\dots,N-k-1$, a one in the row of the a-th entry of $\tilde{\theta}_i(t+j)$ and the column of the b-th entry of $\tilde{\theta}_i(t-k)$, so that $\|\Pi\|=1$ and $\|\Pi\|_F^2 = N-2k$. As the agents are independent, $\tilde{\Theta} := \mathrm{col}(\tilde{\Theta}_1,\dots,\tilde{\Theta}_m)$ is zero-mean Gaussian with a block-diagonal covariance Σ, and $\|\Sigma\|\le c_1$ by (S8), so we may write $\tilde{\Theta} = \Sigma^{1/2}g$ with $g\sim\mathcal{N}(0,I)$. Then

$$\sum_{l=1}^m \omega_l\big(\hat{M}_{j,l}-M_{j,l}\big)_{ab} = g^{\mathsf{T}}Bg-\mathbb{E}[g^{\mathsf{T}}Bg], \qquad B := \tfrac{1}{N-2k}\Sigma^{1/2}\,\mathrm{blkdiag}(\omega_1\Pi,\dots,\omega_m\Pi)\Sigma^{1/2},$$

with $\|B\|\le c_1\|\omega\|/(N-2k)$ and $\|B\|_F\le c_1\|\omega\|/\sqrt{N-2k}$. By the Hanson–Wright inequality (Rudelson and Vershynin, 2013, Theorem 1.1), for an absolute constant $c_0>0$ and all $s\ge 0$,

$$\mathbb{P}\big(|g^{\mathsf{T}}Bg-\mathbb{E}[g^{\mathsf{T}}Bg]|>s\big) \le 2\exp\Big(-c_0\min\Big\{\frac{s^2}{\|B\|_F^2},\frac{s}{\|B\|}\Big\}\Big). \tag{S9}$$

Let $\ell := \log(2p^2/\delta)/c_0$ and $s := c_1\|\omega\|\sqrt{\ell/(N-2k)}$. If $N-2k\ge\ell$, both ratios in (S9) are at least ℓ, so each of the p^2 entries of $\sum_l\omega_l(\hat{M}_{j,l}-M_{j,l})$ exceeds s in absolute value with probability at most $2e^{-c_0\ell}=\delta/p^2$, and a union bound gives $\|\sum_l\omega_l(\hat{M}_{j,l}-M_{j,l})\|\le ps$ with probability at least $1-\delta$. Now let

$$c := p\,c_1\sqrt{\frac{2\big(1+\log_2(2p^2)\big)}{c_0}}, \qquad N_0(\delta) := 4k+\frac{1}{c_0}\log\frac{2p^2}{\delta}. \tag{S10}$$

If $N\ge N_0(\delta)$, then $N-2k\ge N/2$ and $N-2k\ge\ell$, and, as $\delta\le 1/2$, $\log(2p^2/\delta)\le(1+\log_2(2p^2))\log(1/\delta)$, so $ps\le c\|\omega\|\sqrt{\log(1/\delta)/N}$. Thus, for $\delta\in(0,1/2]$ and $N\ge N_0(\delta)$,

with probability at least $1-\delta$,

$$\Big\| \sum_{l=1}^{m} \omega_l \big(\hat{M}_{j,l} - M_{j,l}\big) \Big\| \le c\,\|\omega\| \sqrt{\frac{\log(1/\delta)}{N}}, \tag{S11}$$

which is (16), in the range $\delta \le 1/2$, $N \ge N_0(\delta)$ in which it is used below.

(iii) Consensus error: By (9), $\hat{M}_j^{(i)}(\tau) = \sum_l (W^\tau)_{il} \hat{M}_{j,l}$. The weights $\omega_l^{(i)} := (W^\tau)_{il} - 1/m$ satisfy $\sum_l \omega_l^{(i)} = 0$, as W^τ is row stochastic, and $\|\omega^{(i)}\| \le \|W^\tau - J\| = \|(W-J)^\tau\| \le \sigma^\tau$, as $WJ = JW = J = J^2$. Hence

$$\hat{M}_j^{(i)}(\tau) - \bar{M}_j = \frac{1}{m} \sum_{l=1}^{m} \big(\hat{M}_{j,l} - M_{j,l}\big) + \sum_{l=1}^{m} \omega_l^{(i)} \big(\hat{M}_{j,l} - M_{j,l}\big) + \sum_{l=1}^{m} \omega_l^{(i)} \big(M_{j,l} - \bar{M}_j\big). \tag{S12}$$

Let $\mathcal{E}_\delta$ be the event on which (S11) holds with $\delta/(4m)$ in place of δ, for $j \in \{0,1\}$ and $\omega \in \{\mathbf{1}_m/m, \omega^{(1)}, \dots, \omega^{(m)}\}$. Since $N \ge N_0(\delta/(4m))$, a union bound over these $2(m+1)$ events gives $\mathbb{P}(\mathcal{E}_\delta) \ge 1 - 2(m+1)\delta/(4m) \ge 1-\delta$. On $\mathcal{E}_\delta$, the three terms of (S12) are at most $cm^{-1/2}\sqrt{\log(4m/\delta)/N}$, $c\sigma^\tau\sqrt{\log(4m/\delta)/N}$, and $\|\omega^{(i)}\|_1 D_M \le \sqrt{m}\,\sigma^\tau D_M$, so, by (15),

$$\big\|\hat{M}_j^{(i)}(\tau) - \bar{M}_j\big\| \le \epsilon^{\mathrm{m}}(N,\tau,\delta) \qquad \text{for all } i \in \mathcal{V} \text{ and } j \in \{0,1\}, \text{ on } \mathcal{E}_\delta. \tag{S13}$$

(iv) Bias of the pooled moments: By Section 3, $\bar{M}_1 = A\bar{M}_0 + \bar{\Xi}$. By (S6), (S7), and the Cauchy–Schwarz inequality, $\|\Xi_i\| \le (1+\|A\|)\, b_{k,i} B'_{\theta,i}$, so

$$\|\bar{\Xi}\| \le \epsilon^{\mathrm{b}} := \frac{1+\|A\|}{m} \sum_{i=1}^{m} b_{k,i} B'_{\theta,i}. \tag{S14}$$

(v) Perturbation of the IV estimator: Since $\bar{M}_0 = A^k \bar{\Sigma}_\theta + \bar{\Delta}_0$, Weyl's inequality $\sigma_{\min}(X+Y) \ge \sigma_{\min}(X) - \|Y\|$, $\|\bar{\Delta}_0\| \le \nu_k/2$ and $\epsilon^{\mathrm{m}} \le \nu_k/8$, and (S13) give $\sigma_{\min}(\bar{M}_0) \ge \nu_k/2$ and $\sigma_{\min}(\hat{M}_0^{(i)}(\tau)) \ge 3\nu_k/8$ on $\mathcal{E}_\delta$, where $\nu_k > 0$ as A is invertible and $\bar{\Sigma}_\theta \succ 0$. On $\mathcal{E}_\delta$, writing $\hat{M}_j$ for $\hat{M}_j^{(i)}(\tau)$, the identities $\bar{M}_1\bar{M}_0^{-1} = A + \bar{\Xi}\bar{M}_0^{-1}$ and $X^{-1} - Y^{-1} = -Y^{-1}(X-Y)X^{-1}$ give

$$\hat{A}^{(i)} - A = \big(\hat{M}_1 - \bar{M}_1\big)\hat{M}_0^{-1} - \bar{M}_1\bar{M}_0^{-1}\big(\hat{M}_0 - \bar{M}_0\big)\hat{M}_0^{-1} + \bar{\Xi}\bar{M}_0^{-1}, \tag{S15}$$

so, by (S13) and $\|\bar{M}_1\bar{M}_0^{-1}\| \le \|A\| + 2\epsilon^{\mathrm{b}}/\nu_k$, for all $i \in \mathcal{V}$ on $\mathcal{E}_\delta$,

$$\|\hat{A}^{(i)} - A\| \le \frac{8}{3\nu_k}\Big(1 + \|A\| + \frac{2\epsilon^{\mathrm{b}}}{\nu_k}\Big)\epsilon^{\mathrm{m}}(N,\tau,\delta) + \frac{2\epsilon^{\mathrm{b}}}{\nu_k}. \tag{S16}$$

By (15) and $m^{-1/2} + \sigma^\tau = (1+\sqrt{m}\sigma^\tau)/\sqrt{m}$, the first term gives the finite-sample and heterogeneity terms of Theorem 2 with $C_A := \frac{8c}{3\nu_k}\big(1 + \|A\| + \frac{2\epsilon^{\mathrm{b}}}{\nu_k}\big)$ and $C_W := C_A/c$. The second term is the bias residual: by (S14), (S6), and $\frac{1}{m}\sum_i \sqrt{\operatorname{tr} Q_i} \le \sqrt{\operatorname{tr} \bar{Q}}$ (concavity of the square root),

$$\begin{aligned} \frac{2\epsilon^{\mathrm{b}}}{\nu_k} &\le C_B\big(k\|A - I\| + \sqrt{k \operatorname{tr} \bar{Q}}\big), \\ C_B &:= \frac{2(1+\|A\|)C_\rho}{\nu_k} \max_{i \in \mathcal{V}} \big(\Gamma_i \chi_i B'_{\theta,i}\big) \max\Big\{\frac{\max_i B_{\theta,i}}{\sqrt{3}}, \frac{1}{\sqrt{2}}\Big\}. \end{aligned} \tag{S17}$$

The constants C_A, C_W, C_B do not depend on N, τ, or δ.

Proof of Lemma 3: Let $X(t)$, $S(t)$, $G(t) \in \mathbb{R}^{m\times n}$ be the matrices with i-th rows $x_i(t)^\mathsf{T}$, $s_i(t)^\mathsf{T}$, and $\gamma_i(t)^\mathsf{T}$, where $\gamma_i(t) = \nabla_x \hat{f}_i(x_i(t), t)$ for $t \ge N$, and let $\bar{g}(t) := \frac{1}{m}\sum_i \gamma_i(t)$. For

$t \geq N$ the forecasts are fixed functions of the data, and (11) is the deterministic recursion

$$X(t+1) = W\big(X(t) - \alpha S(t)\big), \qquad S(t+1) = W\big(S(t) + G(t+1) - G(t)\big). \tag{S18}$$

(i) Tracking identity. Multiplying (S18) by $\frac{1}{m}\mathbf{1}_m^\top$ gives $\bar{x}(t+1) = \bar{x}(t) - \alpha\bar{s}(t)$ and $\bar{s}(t+1) = \bar{s}(t) + \bar{g}(t+1) - \bar{g}(t)$ for all $t \geq 0$. Since $\bar{s}(0) = \bar{g}(0)$, induction gives

$$\bar{s}(t) = \bar{g}(t) = \frac{1}{m}\sum_{i=1}^{m} \nabla_x \hat{f}_i(x_i(t), t) \qquad \text{and} \qquad \bar{x}(t+1) = \bar{x}(t) - \alpha\bar{g}(t), \qquad t \geq N. \tag{S19}$$

(ii) Iterate disagreement. By (S18), $(I - J)W = W - J$, and $(W - J)\mathbf{1}_m = 0$, so

$$a_{t+1} \leq \sigma\,(a_t + \alpha c_t). \tag{S20}$$

(iii) Optimization error. Write $\bar{g}(t) = \frac{1}{m}\nabla_x \hat{F}(\bar{x}(t), t) + r_t$, where, by L-smoothness and the Cauchy–Schwarz inequality,

$$r_t := \frac{1}{m}\sum_{i=1}^{m}\big(\nabla_x \hat{f}_i(x_i(t), t) - \nabla_x \hat{f}_i(\bar{x}(t), t)\big), \qquad \|r_t\| \leq \frac{L}{m}\sum_{i=1}^{m}\|x_i(t) - \bar{x}(t)\| \leq \frac{L}{\sqrt{m}}\,a_t.$$

By (S19), $\bar{x}(t+1) = T_t(\bar{x}(t)) - \alpha r_t$, and by (4), $\|\hat{x}^*(t+1) - \hat{x}^*(t)\| \leq L_* d_t$. Therefore $\|\bar{x}(t+1) - \hat{x}^*(t+1)\| \leq (1 - \frac{\alpha\mu}{m})\|\bar{x}(t) - \hat{x}^*(t)\| + \alpha\|r_t\| + L_* d_t$, that is,

$$b_{t+1} \leq \alpha L\, a_t + \big(1 - \tfrac{\alpha\mu}{m}\big) b_t + \sqrt{m}\, L_*\, d_t. \tag{S21}$$

(iv) Tracker disagreement: By (S18) and $(W-I)\mathbf{1}_m = 0$, $X(t+1) - X(t) = (W-I)\big(X(t) - \mathbf{1}_m\bar{x}(t)^\top\big) - \alpha W S(t)$, so $\|X(t+1) - X(t)\|_F \leq 2a_t + \alpha\|S(t)\|_F \leq 2a_t + \alpha\big(c_t + \sqrt{m}\|\bar{s}(t)\|\big)$, where, by (S19), $\nabla_x \hat{F}(\hat{x}^*(t), t) = 0$, and the mL-smoothness of $\hat{F}(\cdot, t)$, $\|\bar{s}(t)\| \leq \frac{1}{m}\|\nabla_x \hat{F}(\bar{x}(t), t) - \nabla_x \hat{F}(\hat{x}^*(t), t)\| + \|r_t\| \leq \frac{L}{\sqrt{m}}(a_t + b_t)$. Collecting,

$$c_{t+1} \leq \sigma\Big(L(2 + \alpha L)\, a_t + \alpha L^2\, b_t + (1 + \alpha L)\, c_t + \Gamma\, d_t\Big). \tag{S22}$$

(v) Spectral radius: Inequalities (S20)–(S22) gives $e_{t+1} \leq \Lambda e_t + v\, d_t$ for $t \geq N$, with

$$\Lambda := \begin{bmatrix} \sigma & 0 & \alpha\sigma \\ \alpha L & 1 - \alpha\mu/m & 0 \\ \sigma L(2 + \alpha L) & \sigma\alpha L^2 & \sigma(1 + \alpha L) \end{bmatrix}, \qquad v := \begin{bmatrix} 0 \\ \sqrt{m}\, L_* \\ \sigma\Gamma \end{bmatrix}, \tag{S23}$$

both nonnegative, since $\alpha\mu/m \leq \mu^2/\big(m(mL + \mu)L\big) < 1$ by the step-size condition.

(vi) Conclusion: Since Λ and v are nonnegative, unrolling the recursion from $t = N$ gives $0 \leq e_t \leq \Lambda^{t-N} e_N + \sum_{q=N}^{t-1} \Lambda^{t-1-q} v\, d_q$. Fix $\rho_{\mathrm{gt}} \in (\rho(\Lambda), 1)$ and $c_\Lambda := \sup_{j \geq 0}\|\Lambda^j\|/\rho_{\mathrm{gt}}^j < \infty$,

$$\|e_t\| \leq c_\Lambda\Big(\rho_{\mathrm{gt}}^{t-N}\|e_N\| + \|v\| \sum_{q=N}^{t-1} \rho_{\mathrm{gt}}^{t-1-q} d_q\Big).$$

Finally, $\|x_i(t) - \hat{x}^*(t)\| \leq \|x_i(t) - \bar{x}(t)\| + \|\bar{x}(t) - \hat{x}^*(t)\| \leq a_t + b_t/\sqrt{m} \leq \sqrt{2}\,\|e_t\|$, which is (18) with $c_{\mathrm{gt}} := \sqrt{2}\, c_\Lambda \max\{1, \|v\|\}$.

Proof of Theorem 4: Fix $t \geq N$ and let $H := t - (N - k)$, so that $\hat{\theta}_i(t) = (\hat{A}^{(i)})^H \tilde{\theta}_i(N - k)$ by (10). Since $\theta(t), \hat{\theta}(t) \in \Theta$, the triangle inequality and (4) give

$$\|x_i(t) - x^*(t)\| \leq \|x_i(t) - \hat{x}^*(t)\| + L_* \sum_{j=1}^{m} \|\hat{\theta}_j(t) - \theta_j(t)\|. \tag{S24}$$

(i) Powers of the identified dynamics: On $\mathcal{E}_\delta$, $\|\hat{A}^{(i)} - A\| \le \epsilon^{\mathrm{id}}(\delta)$ by Theorem 2. Expanding $(\hat{A}^{(i)})^j = \big(A + (\hat{A}^{(i)} - A)\big)^j$, each of the $\binom{j}{l}$ products with l factors $\hat{A}^{(i)} - A$ has norm at most $C_\rho^{l+1}\rho_A^{j-l}\epsilon^{\mathrm{id}}(\delta)^l$, as $C_\rho \ge 1$. Summing, and using $(\hat{A}^{(i)})^H - A^H = \sum_{l=0}^{H-1}(\hat{A}^{(i)})^l(\hat{A}^{(i)} - A)A^{H-1-l}$ with $\rho_A \le \hat{\rho}$, we get on $\mathcal{E}_\delta$

$$\|(\hat{A}^{(i)})^j\| \le C_\rho\big(\rho_A + C_\rho\epsilon^{\mathrm{id}}(\delta)\big)^j = C_\rho\hat{\rho}^j, \qquad \|(\hat{A}^{(i)})^H - A^H\| \le C_\rho^2 H\hat{\rho}^{H-1}\epsilon^{\mathrm{id}}(\delta). \tag{S25}$$

(ii) Term (iii) of (19) (the forecast error): By (S25) and (14), and the independence of the zero-mean process noises, the three terms of (13) satisfy

$$\mathbb{E}_\delta\big\|\big((\hat{A}^{(i)})^H - A^H\big)\tilde{\theta}_i(N-k)\big\| \le C_\rho^2 H\hat{\rho}^{H-1}\epsilon^{\mathrm{id}}(\delta)\, B'_{\theta,i},$$
$$\mathbb{E}_\delta\|A^H\tilde{e}_i\| \le C_\rho\rho_A^H\,\epsilon_i^{\mathrm{w}},$$
$$\mathbb{E}_\delta\Big\|\sum_{j<H}A^j w_i(t-1-j)\Big\| \le \Big(\operatorname{tr}\sum_{j<H}A^j Q_i(A^j)^\mathsf{T}\Big)^{1/2} = \sigma_{H,i},$$

and $\sigma_{H,i}^2 \le \operatorname{tr}\Sigma_{\theta,i}$ since $\Sigma_{\theta,i} = \sum_{l\ge 0}A^l Q_i(A^l)^\mathsf{T}$. Hence $\mathbb{E}_\delta\|\hat{\theta}_i(t) - \theta_i(t)\| \le \epsilon_i^{\mathrm{f}}(N,t;\delta)$ for every $i \in \mathcal{V}$, giving term (iii) of (19).

(iii) Terms (i) and (ii) of (19) (the tracking error): By (10), $\hat{\theta}_j(q+1) - \hat{\theta}_j(q) = (\hat{A}^{(j)})^{q-N+k}(\hat{A}^{(j)} - I)\tilde{\theta}_j(N-k)$, so, on $\mathcal{E}_\delta$, (S25) and $\|\hat{A}^{(j)} - I\| \le \|A - I\| + \epsilon^{\mathrm{id}}(\delta)$ give, for $q \ge N$,

$$d_q \le \sum_{j=1}^m\|\hat{\theta}_j(q+1) - \hat{\theta}_j(q)\| \le C_\rho\hat{\rho}^{q-N+k}\big(\|A - I\| + \epsilon^{\mathrm{id}}(\delta)\big)\sum_{j=1}^m\|\tilde{\theta}_j(N-k)\|.$$

As $\rho_{\mathrm{gt}}, \hat{\rho} \le \rho_\star$, $\sum_{q=N}^{t-1}\rho_{\mathrm{gt}}^{t-1-q}\hat{\rho}^{q-N+k} \le \hat{\rho}^k(t-N)\rho_\star^{t-N-1}$. Inserting this in (18), which holds for every realization of the data, and taking $\mathbb{E}_\delta$ with $\mathbb{E}\|\tilde{\theta}_j(N-k)\| \le B'_{\theta,j}$ gives

$$\mathbb{E}_\delta\|x_i(t) - \hat{x}^*(t)\| \le c_{\mathrm{gt}}\rho_{\mathrm{gt}}^{t-N}\mathbb{E}_\delta\|e_N\| + c_{\mathrm{gt}}C_\rho\hat{\rho}^k(t-N)\rho_\star^{t-N-1}U_N,$$

which are terms (i) and (ii).

References

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